

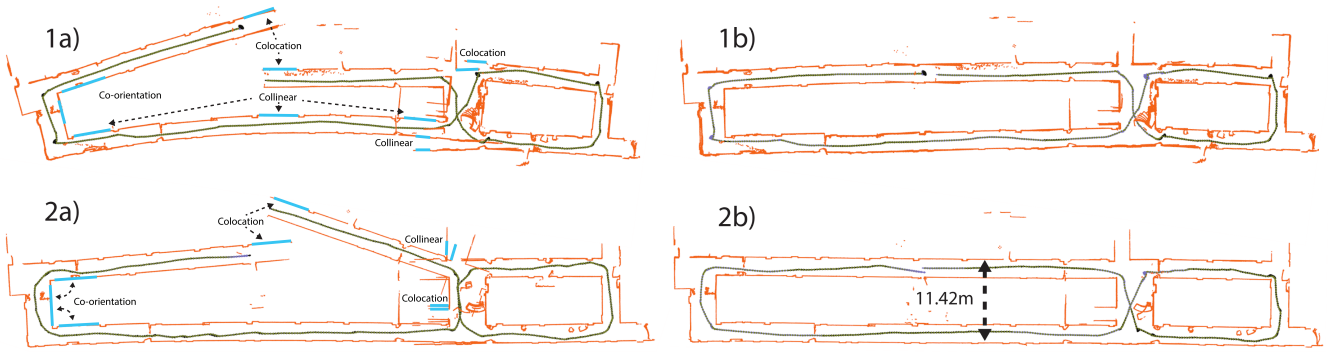


Figure 6: Initial and final maps from HitL-SLAM. Each map is of the same floor, and consists of between 600 and 700 poses. Maps in the left column (a) are initial maps, and maps in the right column (b) are final maps. Observations are shown in orange and poses are shown as arrows. Poses which are part of a human constraint are blue, while those which are not are in black.

we have ground truth dimensions. This creates sequences of “lost” poses throughout the pose-chain which rely purely on odometry to localize, thus accruing error over time. We then impose human constraints on the resultant map and compare to ground truth, shown in Fig. 8. Note that the human corrections do not directly enforce any of the measured dimensions. The initial map shows a room width of 5.97m, and an angle between opposite walls of  $4.1^\circ$ . HitL-SLAM finds a room width of 6.31m, while the ground truth width is 6.33m, and produces opposite walls which are within  $1^\circ$  of parallel. Note also that due to the limited sensor range, the global correctness must come from proper application of human constraints to the factor graph including the “lost” poses between wall observations.

To quantitatively evaluate accuracy on larger maps, where exact ground truth is sparse, we measured an inter-corridor spacing (Fig. 6 2b), which is constant along the length of the building. We also measured angles between walls we know to be parallel or perpendicular. The results for ground truth comparison before and after HitL-SLAM, displayed in Table 1, show that HitL-SLAM is able to consistently reduce the map errors even when given poor quality initial observations from pose  $x_i$  show as free space and observations from pose  $x_j$  show as occupied space. We define the total inconsistency  $\mathbf{I}$  over the map as the pair-wise sum of inconsistencies between all pairs of poses,  $\mathbf{I} = \sum_{i=1}^{N-1} \sum_{j=i+1}^N I_{i,j}$ . The inconsistency metric thus serves as a quantitative metric of global registration error between observations in the map, and allows us to track the effectiveness of global registration using HitL-SLAM over multiple iterations. The initial inconsistency values for maps LGRC 3A and 3B were  $297.5m^2$  and  $184.3m^2$ , respectively. The final inconsistency values were  $47.6m^2$  and  $3.7m^2$ , respectively, for an average inconsistency reduction of 91% relative to the initial map, thus demonstrating the improved global consistency of the map generated using HitL-SLAM. Fig. 6 and Fig. 9 offer some qualitative examples of HitL-SLAM’s performance.

To test the scalability of HitL-SLAM, we gathered several datasets with between 600 and 700 poses, and one with over 3000 poses and nearly 1km of indoor travel between three large buildings. Fig. 6 shows some of the moderately sized

| Map        | Samples |    | Input Err.    |      | HitL-SLAM Err. |      |
|------------|---------|----|---------------|------|----------------|------|
|            | A       | T  | A( $^\circ$ ) | T(m) | A( $^\circ$ )  | T(m) |
| Lost Poses | 10      | 4  | 3.1           | 0.07 | 1.0            | 0.02 |
| LGRC 3A    | 14      | 10 | 9.8           | 3.3  | 1.5            | 0.06 |
| LGRC 3B    | 14      | 10 | 7.6           | 3.1  | 1.1            | 0.02 |
| BIG MAP    | 22      | 10 | 5.9           | 2.8  | 1.6            | 0.03 |
| Mean       | 60      | 34 | 6.74          | 2.71 | 1.4            | 0.04 |

Table 1: Quantitative mapping errors using HitL-SLAM compared to ground truth, in the input maps, and after HitL-SLAM. The ‘Samples’ column denotes how many pairwise feature comparisons were made on the map and then compared to hand-measured ground truth. Angular (A) errors are in degrees, translation (T) errors in meters.

maps, and Fig. 7 details the largest map. 16 constraints were required to fix the largest map, and computation time never exceeded the time required to re-display the map, or for the human to move to a new map location.

All maps shown in Fig. 6 were corrected interactively by the human using HitL-SLAM in under 15 minutes. Furthermore, HitL-SLAM solves two common problems which are difficult or impossible to solve via re-deployment: 1) a severely bent hallway, in Fig. 6 1a), and 2) a sensor failure, in Fig. 6 2a) which caused the robot to incorrectly estimate its heading by roughly 30 degrees at one point. Combined, these results show that incorporating human input into metric mapping can be done in a principled, computationally tractable manner, which allows us to fix metric mapping consistency errors in less time and with higher accuracy than previously possible, given a small amount of human input.

## Conclusion

We present Human-in-the-Loop SLAM (HitL-SLAM), an algorithm designed to leverage human ability and meta-knowledge as they relate to the data association problem for robotic mapping. HitL-SLAM contributes a generalized framework for interpreting human input using the EM algorithm, as well as a factor graph based algorithm for incorporating human input into pose-graph SLAM. Future work in

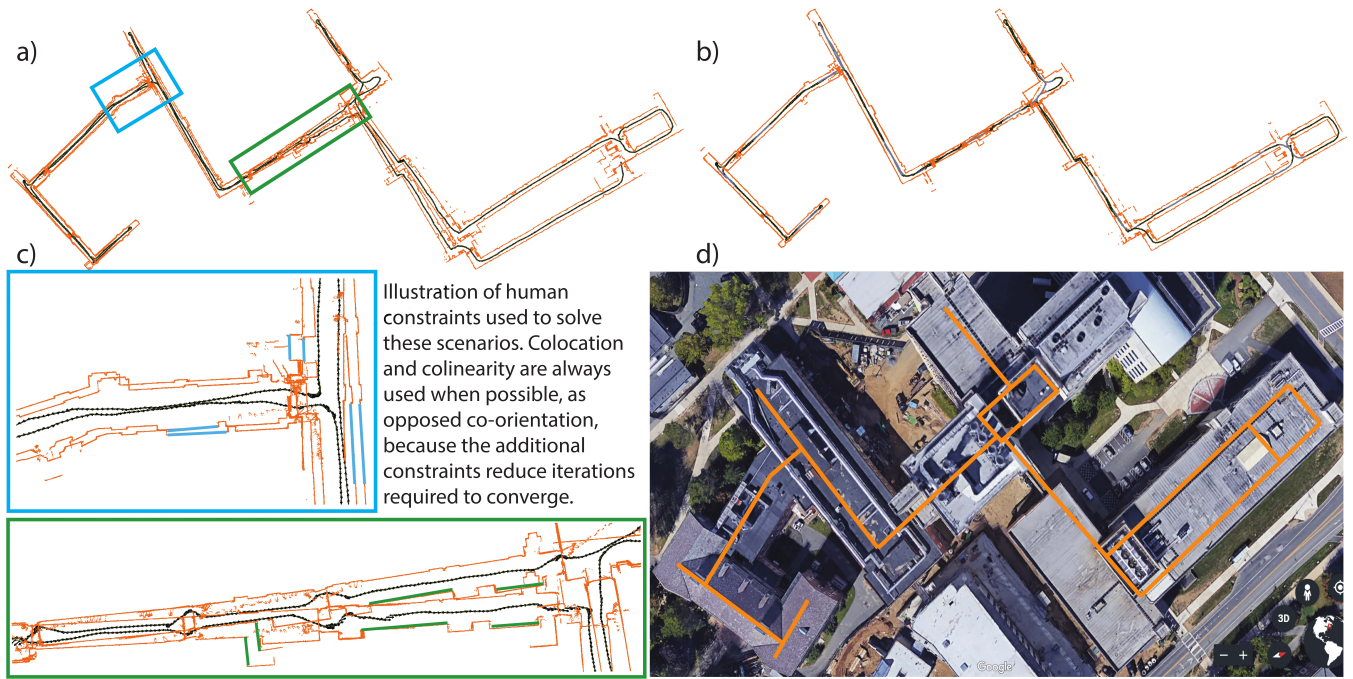


Figure 7: A large map a) corrected by HitL-SLAM b) using human correction, some of which are highlighted c). d) shows an approximate overlay of the map onto an aerial image of the complex from google earth. The map contains over 3000 poses.

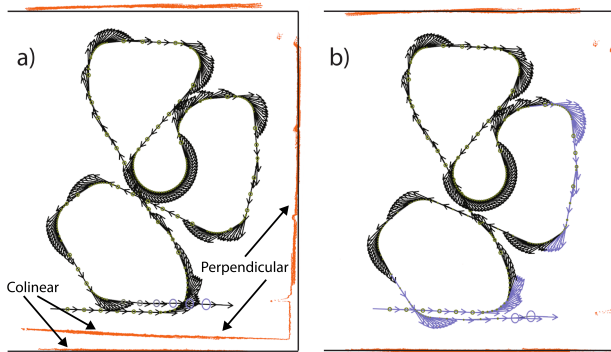


Figure 8: Initial a) and final b) maps for the 'lost poses' experiment. Observations are shown in orange, poses are black arrows, and ground truth (walls) is represented by the black lines. Poses involved in human constraints are colored blue.

this area could proceed towards further reducing the human requirements, and extending this method for higher dimensional SLAM and for different sensor types.

### Acknowledgments

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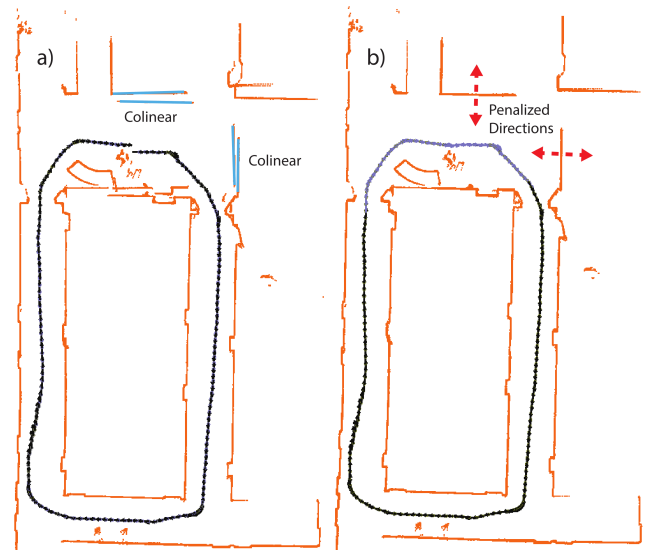


Figure 9: Example of how rank-deficient (collinear) constraints can be combined to effect a full rank constraint. Often, it is easier to tell if two sets of observations are collinear rather than colocated. Supporting rank-deficient constraints allows HitL-SLAM to operate on a larger set of maps.

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